GenFSDiff

Few-shot steered diffusion for satellite-telemetry anomalies

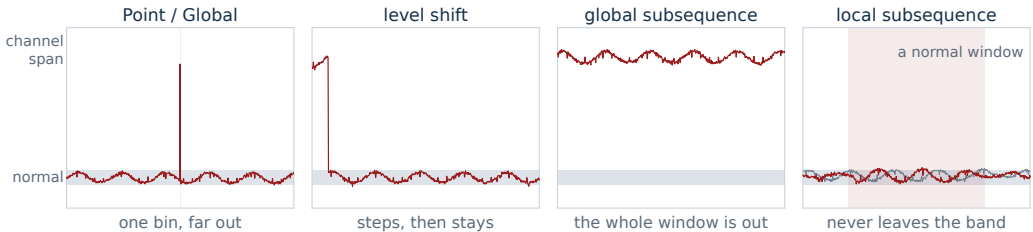
ESA-ADB Mission 1, channels 41–46

- **Problem.** Seven years of telemetry per channel hold only **22** labelled anomaly events.
  Too few anomalies to train a reasonable detector.
- **Approach.** A generative model to generate an anomaly of a chosen kind (+size and position).
- **Result.** A detector trained on real anomalies plus generated ones improves its performance.

14 October 2026

The data: ESA-ADB Mission 1

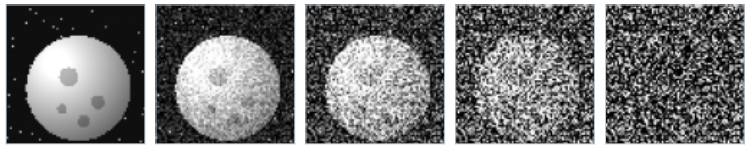
the four kinds we generate	what it looks like	windows	events
Point / Global	one sample far outside the normal range	18	3
level shift	the signal steps to a new level and stays	6	1
global subsequence	a long stretch outside the normal range	636	17
local subsequence	a stretch of normal values, wrong for the context	12	2



What are diffusion models?

The framework behind e.g. image generators; we apply it to telemetry

reverse: a trained neural network ε_θ removes it one step at a time



x_0 data $\qquad$ x_T pure noise

forward: add Gaussian noise — fixed, nothing is learned

Forward process $q(x_t \mid x_0)$, model's guess of the original $\hat{x}_0$:

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t) I\big)$$

$$\hat{x}_0 = \big(x_t - \sqrt{1-\bar{\alpha}_t}\, \varepsilon_\theta(x_t, t)\big) / \sqrt{\bar{\alpha}_t}$$

- $\bar{\alpha}_t$: a fixed schedule — how much of the original signal is left at step t.
- ε_θ: trained to predict the noise that was added.

How a (nominal) sample is generated

Start from noise, predict the added noise, and remove it. Repeat.

The loop, as implemented

$x_T \sim \mathcal{N}(0,\, I)$ — draw pure noise

for $t = T-1,\, \ldots,\, 0$:

- $\varepsilon \leftarrow \varepsilon_\theta(x_t,\, t,\, c)$ — one network call: the noise it sees in x_t
- $\hat{x}_0 \leftarrow \left(x_t - \sqrt{1-\bar{\alpha}_t}\,\varepsilon\right) \big/ \sqrt{\bar{\alpha}_t}$ — undo the forward process: guess of x_0
- $x_{t-1} \leftarrow \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\varepsilon$ — *same* noise back, at level $t-1$
- $t \leftarrow t-1$ — go to the next step

return x_0

Our setting: Steer the sample to generate an anomaly

One sampling step with the steering edit E

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,(\hat{x}_0 - \mathbf{E}) + \sqrt{1-\bar{\alpha}_{t-1}}\,\varepsilon$$

E is the only thing we add — the denoiser itself is never retrained.

E should have the following effects:

- make the sample unusual enough to be an anomaly
- give it the shape and size of the anomaly kind we asked for

Now, what is **E**?

How we steer

Two forces, applied to the model's running guess $\hat{x}_0$ at every step

1. Be as unusual as normal data ever gets — the *shell*

$$h(x) = -\tau \log \sum_{r \in R} e^{-\|e(x) - e(r)\|^2 / \tau}$$ how unusual x is

$Q =$ 99th percentile of h, held-out nominal

$$\text{shell} = \{\, x \;:\; |h(x) - Q| \le \delta \,\}$$

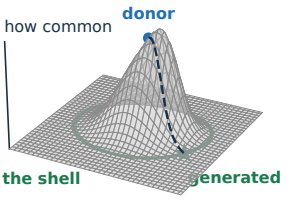
2. Have the shape and size we asked for

$$\langle w,\, x \rangle \;=\; \sum_t w[t]\, x[t] \;\longrightarrow\; a, \qquad \sum_t w[t] = 0$$

where w encodes the shape and a the size

Together, the edit E

$$\mathbf{E} \;=\; \underbrace{\lambda\, \nabla_{\hat{x}_0} \big(h(\hat{x}_0) - Q\big)^2}_{\text{onto the shell}} \;+\; \underbrace{\big(\langle w, \hat{x}_0 \rangle - a\big)\, w \,/\, \|w\|^2}_{\text{to the asked-for shape}}$$

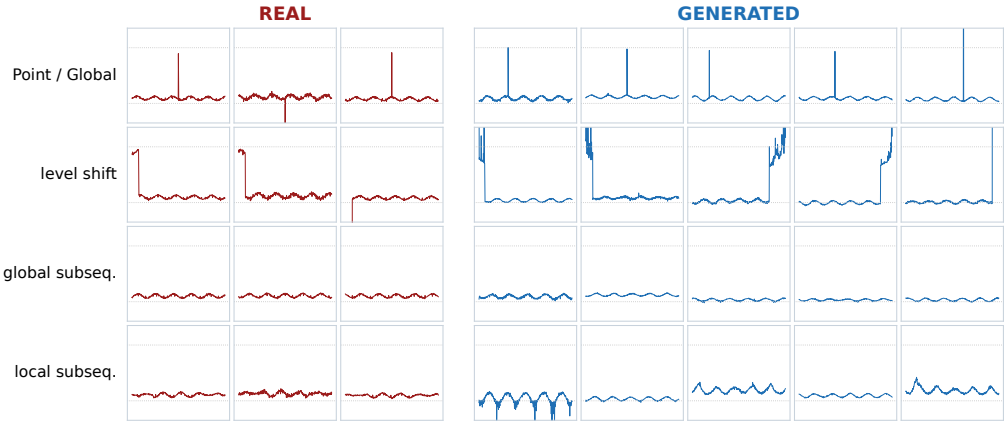


In words. Make the window as unusual as the most unusual 1% of normal telemetry

$e(\cdot)$ is a trained encoder.

What the model produces

three real windows and five generated ones, per kind



Realism and diversity

Evaluation against other existing generators, on the two metrics the GenIAS paper reports

ARP — realism

How close the *real* anomalies are to their nearest generated window:

$$\mathrm{ARP} = \frac{1}{1 + \frac{1}{|A|}\sum_{a \in A} \min_{g \in G} \|\phi(a) - \phi(g)\|_2}$$

EDI — diversity

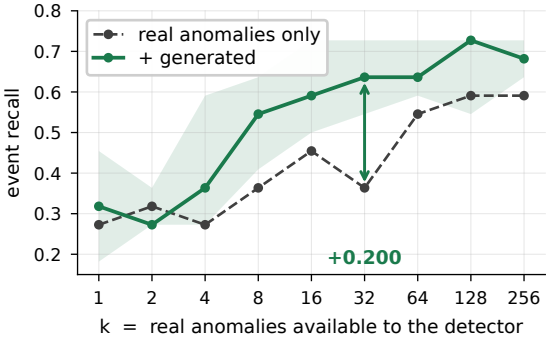
Entropy-like term of the distribution of the generated windows:

$$\mathrm{EDI} = -\sum_{j=1}^{k} p_j \log p_j$$

	ARP	EDI
real anomalies (target)	0.792	1.48
ours	**0.509**	**2.44**
GenIAS	0.317	1.14
CutAddPaste	0.323	1.27
handcrafted injection	0.319	1.31

Does it help a detector?

k real anomalies plus *k* generated ones, against *k* real alone. recall at 1% false-alarm rate



k	real only	+ generated	gain
8	0.364	0.545	**+0.155***
16	0.455	0.591	+0.164
32	0.364	0.636	**+0.200***
64	0.545	0.636	**+0.136***
128	0.591	0.727	+0.118
256	0.591	0.682	+0.109

* 95% confidence interval excludes zero

+0.200 event recall at $k = 32$
4.4 more events of 22 detected

- **Comparison against existing generators at $k = 256$: GenFSDiff 0.682 vs GenIAS 0.591, CutAddPaste 0.545.**

The detector: a 31k-parameter 1-D CNN on the raw window — three convolutions with SiLU activations ($1 \to 32$ width 7, $32 \to 64$ width 5, $64 \to 64$ width 5, each stride 2) **Held-out anomalies:** 22 anomaly events

Summary, and what is not settled

What we showed

- GenFSDiff generates synthetic anomalies of a *chosen* kind and size, without retraining the denoiser.
- Adding them to training improves anomaly detection on ESA-ADB Mission 1.

What is not settled

- **Continuous channels only.** Discrete telemetry (status flags, modes) would need e.g. discrete diffusion.
- **One channel at a time.** Anomalies that live in the relation between channels are not generated. Potential to include causality here?